

The Blend Wall: India's Ethanol Programme Has Peaked at E20

Thematic Equity Research | Indian Automotive Sector

Executive Summary

India's ethanol blending programme has achieved something genuinely rare: a government target beaten by six years — blending crossed 20% in October 2025, E20 became the standard petrol grade from April 2026, and specifications for E22 through E30 are already notified. The question for investors is whether this momentum can continue, or whether E20 is, for all practical purposes, the destination.

We believe it is the destination. The headline trajectory looks compelling — 1.4% blending in 2014 to 20% in 2025, regulatory scaffolding in place for higher blends, and India's 88% crude import dependence giving the programme a durable energy-security justification. But the three structural pillars that made E20 achievable have either eroded or inverted, and the new foundations required for higher blends are not yet in place.

The first is economics. The programme's early rationale was almost elegant: a chronic sugar surplus needed clearing, mills needed to pay cane farmers, and ethanol provided a guaranteed offtake at administered prices. That architecture is gone. Maize is now the dominant feedstock — nearly half of all supply — and the government itself acknowledged in August 2025 that the weighted average ethanol procurement price of ₹71.32 per litre now exceeds the cost of refined petrol before taxes. NITI Aayog's 2021 roadmap recommended E20 be sold at a lower retail price to compensate consumers for the calorific penalty. That recommendation was never implemented. Indian consumers today pay the same or more for a fuel that delivers roughly 13% less energy per litre. The transfer has reversed — consumers are now subsidising the ethanol industry, not the other way around.

The second pillar is feedstock. When the programme ran on molasses — a genuine by-product of sugar production — it had a logical coherence: convert waste to fuel, cut imports, support farmers. The feedstock base today looks nothing like that. Maize diversion to ethanol rose roughly sixteen-fold in two years; India has since turned net maize importer; prices climbed from ₹14,000 per tonne to ₹24,000–25,000; and the poultry and dairy industries — which account for over 60% of domestic maize demand — have been visibly squeezed. The programme that began as a solution to a sugar surplus has become a

contributor to food-price stress. That contradiction does not disappear as blending rises. It compounds.

The third is fleet compatibility. E20 sits at the practical edge of what a legacy ICE vehicle — designed for E5, updated in rubber compounds from April 2023 — can tolerate without meaningful efficiency loss or seal degradation. Higher blends require dedicated flex-fuel engines. Those vehicles are now launching: Maruti's Wagon R FFV arrived in June 2026; Hero launched flex-fuel motorcycles days earlier. But flex-fuel vehicles face an 18–40% GST rate against 5% for EVs — a fiscal architecture almost designed to ensure they remain government showcases rather than consumer choices. And the E85 retail infrastructure that flex-fuel vehicles would actually run on is essentially nonexistent.

The global precedent should settle the question. Every market that has attempted to push materially beyond E20 has eventually retreated or stalled. The United States — the world's largest ethanol producer — is de facto stuck at E10; compelling E15 and E85 uptake costs an estimated \$770 per incremental gallon. China announced a nationwide E10 mandate in 2017 and quietly abandoned it before achieving it. Thailand built the most ambitious gasohol programme in Southeast Asia, reached E85, and reversed — PTT and OR discontinued E85 sales from February 2026, consolidating on E20 and redirecting surplus ethanol to Sustainable Aviation Fuel. The EU has capped food-crop biofuels at 7% and bet on electrification. Only Brazil has genuinely broken through — with roughly 7,000 litres of ethanol per hectare, an 8–10 times sugarcane energy balance, 90% of its car fleet capable of running on any blend, and hydrous E100 priced at the pump at roughly 39% below gasoline. Brazil's endowments of land, water, yield, and two decades of flex-fleet investment are not transferable. India does not get to borrow the conclusion without inheriting the conditions.

Our verdict is that E20 is India's durable medium-term outcome — not a waypoint on the road to something higher. A slow, administered grind toward E25–E27 by the end of the decade is the most likely path. E85 and E100 at mass scale within ten years is not. For the auto industry, the implication is direct: the structural winner is the E20-calibrated ICE and the strong hybrid, not the flex-fuel vehicle the current policy narrative is beginning to describe. And there is one dimension of this story that consistently gets underweighted — even under total success in the petrol pool, ethanol leaves India's diesel pool entirely untouched. Petrol is 40 MMT; diesel is 91 MMT. Solving petrol does not solve India's oil import dependence. It solves, at most, a third of it.

Investment Conclusion

Our thesis is this: India's ethanol blending programme is a genuine policy success story, and E20 is approximately where the success ends. The programme that accomplished in eleven

years what most countries never achieve at all has now run into the same wall that stopped everyone else. We expect a slow, administered progression toward E25–E27 over the remainder of this decade. We do not expect E85 or E100 to achieve mass adoption in India within ten years, and we think the policy narrative currently building around flex-fuel vehicles significantly overstates the commercial opportunity.

The single most important reason for this view is not engineering or economics — it is the global record. No advanced market has sustained ethanol blends above E20 except Brazil, and Brazil's success rests on structural endowments that India does not possess: yields, water, land, energy balance, two decades of flex-fleet construction, and consumer fuel choice at the pump. The US is stuck at E10. China abandoned E10 before achieving it. Thailand built E85 and dismantled it. Europe has capped food-crop biofuels and bet on electrification. If the world's most committed ethanol producer cannot push past E10, India's ambition to reach E85 and E100 should attract a very high burden of proof.

The single most important risk to this thesis is political, not analytical. The government has enormous capital invested in the programme's narrative, and a meaningful rise in crude prices — above \$90–100 per barrel — would sharply improve the economics of higher-blend ethanol. A formal E25–E27 mandate after October 2026, combined with GST rationalisation for flex-fuel vehicles and material progress on E85 retail, would cause us to revisit. We are watching all three signals.

For the automotive sector, this thesis holds through the end of the decade. E20-calibrated ICE and strong hybrids are the structural winners. Flex-fuel is a regulated niche until the GST and infrastructure conditions change. EVs win two-wheelers on total cost of ownership in the medium term regardless of what happens to ethanol. And ethanol, even under total success, does nothing to reduce dependence on the diesel pool — which at 91 MMT is more than double the petrol pool ethanol can address.

The thesis is bounded to the current decade. Beyond 2030, the picture is genuinely uncertain — 2G cellulosic ethanol could improve, crude could structurally rerate, the GST could be rationalised, and a mature flex-fuel fleet could be in place. We are not making a ten-year call against ethanol. We are making a five-year call that the transition narrative materially outruns the commercial reality.

1. E20 Is a Genuine Achievement — Which Is Precisely Why It Matters That the Pillars Beneath It Have Collapsed

Let us start with what India got right. Ethanol blending rose from 1.4% in ESY 2013-14 to roughly 20% in October 2025 — crossing the government's 2030 target six years early, faster than almost any comparable emerging-market energy transition. Over the same

period, installed distillery capacity grew from less than 2 billion litres to roughly 19–20 billion litres across 499 distilleries. OMC ethanol offtake crossed 1,000 crore litres for the first time in ESY 2024–25. From April 1, 2026, E20 with a minimum RON 95 became the standard petrol grade nationwide — E0 effectively phased out, BIS specifications for E22 through E30 already notified under IS 19850:2026, and Finance Ministry excise exemptions for those grades granted in June 2026. By any reasonable measure, this is an impressive programme execution.

The danger is in mistaking what enabled that execution for what would be needed to go further.

The Three Pillars, and Why They Have Eroded

E20 was built on three things that are no longer available in the same form. The first was a genuine supply surplus. The programme's original logic — converting molasses, a sugar by-product that mills paid to dispose of, into blendable fuel — gave it an almost self-funding coherence. There was no food-security tension; the feedstock was waste. There was no contested land use; the sugarcane was already being grown. Between ESY 2013–14 and ESY 2020–21, molasses and cane-juice-based ethanol dominated supply, and the programme was essentially solving a waste disposal problem at scale. That feedstock base is now a minority share — just 31% of supply — and the shift to grain tells a materially different story, which we address in Chapter 3.

The second pillar was administered economics that, until recently, actually worked. The government set ex-mill prices for ethanol above sugar-equivalent margins, creating a genuine incentive for mills to divert cane to ethanol rather than sugar. OMCs were mandated to buy. Interest subvention of 6% was provided for distillery investment. GST on ethanol was cut from 18% to 5% in 2022. The result was a capacity build-out of over ₹40,000–50,000 crore. What those same economics look like today — with procurement prices above petrol's pre-tax base — is a question the programme has not yet answered publicly.

The third was fleet headroom. E10, and then E20, could be introduced into a legacy fleet with modest material upgrades — rubber compounds, revised ECU calibration — phased in at OEM level from April 2023. Consumers could not opt out, but the disruption to older vehicles was officially minor. That headroom is now consumed. The next step requires not a quiet OEM compliance update but a structural change in the vehicle the consumer actually buys: a flex-fuel engine at 18–40% GST, capable of running E85, in a retail network for E85 that does not yet exist.

The Constraint Has Flipped

As Exhibit 1 would show — blending percentage against installed distillery capacity from 2014 to 2026 — the capacity curve and the blend curve have now decisively decoupled. For ESY 2025–26, OMCs tendered approximately 1,050 crore litres but received offers of

roughly 1,776 crore litres — more than 1.7 times the requirement. Installed capacity is approximately 1,822 crore litres against E20 demand of roughly 1,050 crore litres. The All India Distillers' Association reports utilization of just 25–30%, with fresh approvals for new plants halted. The constraint has flipped from supply-limited to demand-limited. The lobby pushing for E25, E30, and even E50 is doing so not because India needs more blending to meet a fuel demand that cannot be met otherwise — it is doing so because it has built factories that have nothing to do. That is a critical context for reading every industry statement about higher-blend ambition.

2. The Economics Have Inverted — And No One Is Telling the Consumer

There is a sentence buried in a Ministry of Petroleum release from August 2025 that deserves more attention than it has received. MoPNG acknowledged that the weighted average ethanol procurement cost — at ₹71.32 per litre as of July 31, 2025, inclusive of transport and GST — now exceeds the cost of refined petrol. Let that land: ethanol, the fuel the government has mandated as a cheaper, domestically produced substitute for imported petrol, now costs more than the petrol it is displacing, before retail taxes.

This is not a minor detail. It is the central economic fact of the programme as it currently stands.

How the Inversion Happened

The inversion happened for a straightforward reason. Maize-based ethanol, the fastest-growing feedstock category, carries a procurement price of ₹71.86 per litre — compared with ₹52.92 in ESY 2021-22. That rise reflects the maize price inflation the programme itself caused; as the industry consumed more maize, prices rose from ₹14,000–15,000 per tonne to ₹24,000–25,000, and those costs fed directly into grain ethanol economics. Even C-heavy molasses ethanol — historically the cheapest input — now sits at ₹57.97 per litre. The feedstock that was once a waste product clearing a surplus now has to compete with food uses that have become expensive.

Exhibit 2 — ethanol procurement price by feedstock against petrol's pre-tax base, ESY 2021-22 to ESY 2024-25 — would make this crossover visible. The lines cross in approximately ESY 2023-24, and maize ethanol has not looked back.

The Recommendation That Was Never Implemented

NITI Aayog's 2021 Roadmap for Ethanol Blending was explicit on what should follow from the calorific reality: "the retail price of such fuels should be lower than normal petrol to

compensate for the reduction in calorific value." The calorific reduction is real — ethanol has approximately 65% of petrol's volumetric energy, meaning E20 carries roughly 13% less energy per litre than E0. The recommendation was never implemented. Indian consumers today pay the same pump price — or occasionally more — for fuel with meaningfully less range per litre. They have no way to opt out; E0 is effectively gone from the forecourt. Brazil solved this through price parity at the pump, where hydrous E100 is priced in real time against gasoline and consumers choose based on kilometres per currency unit. India has the mandate without the transparency, and the energy penalty without the consumer compensation.

The Crude Price Question

The sensitivity to crude oil deserves note because it is the one genuine counterargument to the economic case against higher blends. At \$90–100 per barrel crude, administered ethanol economics become considerably more defensible — the macro-hedging rationale strengthens, and the spread between ethanol procurement and petrol cost narrows. At \$50–60 per barrel, the programme is straightforwardly uneconomic and survives only by mandate. Current crude pricing sits in a base-case range where the programme is roughly margin-neutral on a policy-supported basis — viable, but not self-sustaining, and not a particularly strong argument for pushing blends significantly higher. It is worth noting that the government's own benefit claims — forex savings now cited at ₹1.84 lakh crore, crude substituted at 302 lakh MT, CO₂ avoided at 909 lakh MT (Puri, June 2026) — have escalated considerably across successive statements and are self-reported figures not subject to independent audit. An independent estimate of crude-import savings through 2024 puts the figure materially lower. We treat the official numbers as directional indicators, not verified accounting.

3. A Programme Built on Waste Has Become a Drain on Food

The original justification for India's ethanol blending programme was not energy security, not carbon reduction, and not technology development. It was a sugar crisis. Mills in Uttar Pradesh and Maharashtra owed farmers thousands of crores in sugarcane arrears; excess sugar was depressing global prices; and the government needed a mechanism to convert surplus sucrose into something the market would absorb at administered prices. Ethanol was the answer. The programme's environmental and energy-security credentials were genuine but secondary — they were reasons to like it, not reasons it existed.

That founding logic has been completely abandoned.

The Feedstock Migration Is Not Reversible Without a Policy Reversal

Grain now accounts for approximately 69% of ethanol feedstock, with sugarcane fallen to 31%. Maize alone has become the single largest input at roughly 48–50% of supply, after rising from a negligible share as recently as ESY 2020–21. Maize diverted to ethanol rose from 0.8 million tonnes in ESY 2022–23 to approximately 12.7 million tonnes in ESY 2024–25 — roughly a sixteen-fold increase in two years. In ESY 2025–26, rice allocation is targeted to rise to 90 lakh tonnes, up from 52 lakh tonnes the prior year.

Maize is not a waste product. It is the primary input for poultry feed, dairy feed, and starch production — industries that together account for more than 60% of India's domestic maize consumption. The consequence of this diversion was immediate and visible: India, a net maize exporter of 3.7 million tonnes in FY2022, has become a net importer. Domestic maize prices rose by roughly 60–70% over two years. Poultry producers faced input cost inflation that the retail price of eggs and broilers did not fully absorb. DDGS — the by-product of grain ethanol — partially offsets the feed loss but undercuts soymeal in the process, squeezing soybean farmers who had nothing to do with the ethanol programme.

Exhibit 3 — feedstock mix evolution from ESY 2018 to ESY 2026, as a stacked area chart — would make the structural shift unmistakable. The chart would show a programme that began in cane juice and molasses and has quietly re-engineered itself into a grain-processing industry.

The Water Equation Is Quietly Catastrophic

The water dimension amplifies this concern substantially. NITI Aayog's 2021 Roadmap, drawing on University of Twente data, puts water intensity at approximately 2,860 litres per litre of sugarcane ethanol, rising to roughly 4,670 litres for maize-based and 10,790 litres for rice-based ethanol. Sugarcane and paddy combined already use approximately 70% of India's total irrigation water. CSTEP modelling suggests that scaling ethanol demand could raise annual irrigation requirements by roughly 50 billion cubic metres by 2070 — in a country where the government's own Composite Water Management Index warns that 21 major cities, including Delhi, Bengaluru, and Chennai, face severe groundwater depletion by 2030.

The point is not that these numbers are exact. It is that the directional argument is overwhelming, and it runs directly against the programme's stated rationale. If India's ethanol ambition were genuinely about waste-to-value conversion in a water-stressed country, the feedstock mix would be heading toward sugarcane and second-generation cellulosic. It is heading in exactly the opposite direction. The logical conclusion — one the programme has not confronted publicly — is that India's ethanol industry has migrated from clearing a commodity surplus to creating a structural bid for food-security resources that the country can ill afford to divert.

Second-Generation Ethanol Has Not Arrived, and Will Not Arrive Soon

Second-generation cellulosic ethanol was supposed to be the answer to all of this. It was not supposed to need food crops at all — agricultural waste, rice straw, bagasse, and similar feedstocks were meant to provide a pathway to high-blend ethanol without the food-versus-fuel tension. The reality has been consistently disappointing. IOCL's Panipat 2G plant — the flagship — runs at just over 50% of design capacity. Of the twelve 2G plants planned under the PM JI-VAN Yojana scheme, only a handful are operational, producing a combined output of roughly 32 million litres per year against the billions required for material blending. The scheme has been extended to 2028-29 with approximately ₹1,969 crore allocated, but after a decade of subsidised effort, 2G remains a chronic underperformer — hampered by the logistics and economics of agricultural waste collection, higher capital costs, and process complexity that the first-generation industry has never had to face.

This matters for the higher-blend thesis because 2G was always the implicit backstop — the technology that would provide the feedstock runway once first-generation sources were exhausted or politically problematic. That backstop is not ready. The programme pushing toward E30 and beyond will have to either intensify the food-system pressure through more grain diversion, or slow down and wait for a technology that has been perpetually five years away.

4. The Engineering Reality: E20 Is Where the Legacy Fleet Runs Out of Runway

The physics of ethanol blending are not negotiable. Ethanol's lower heating value is approximately 26.8–28.9 MJ per kilogram against roughly 42.5 for gasoline; its volumetric energy density is approximately 21 MJ per litre against roughly 32 — about 65% of petrol's energy content per unit of volume. A litre of E20 therefore carries approximately 7% less volumetric energy than a litre of E0. Because fuel consumption does not rise proportionately — fuel system calibration and octane effects partially compensate — the real-world mileage impact at E20 for a properly calibrated vehicle runs between 1% and 6% per ARAI and NITI estimates, with the lower end applying only to dedicated E20-tuned engines introduced from April 2025 under BS6 Phase 2.

For the legacy fleet — vehicles designed for E5, operating on E20 with only the material compliance updates introduced from April 2023 — the mileage story is more complicated.

The Mileage Gap: Official vs Experienced

NITI's official 1–6% range has been challenged consistently by consumer surveys. A LocalCircles survey released in February 2026, covering 50,000+ respondents across 301 districts, found that approximately one in two owners of vehicles purchased before 2023 reported reduced fuel efficiency; a separate October 2025 wave put the mileage decline

claim at roughly 80%, with around 52% reporting unusual wear or repair needs. These surveys are self-reported — not instrumented, and not controlled for driving pattern changes or maintenance quality — and should not be cited as engineering evidence. But the volume and persistence of the complaints carries its own signal, and the fact that approximately 55% of respondents in these surveys said they wanted E0 or E10 back is a consumer sentiment data point that policymakers will find difficult to ignore indefinitely.

The Supreme Court dismissed a PIL on the matter in September 2025, with the Attorney General characterising the petitioner as a proxy for a competing commercial lobby rather than representing genuine consumer interest. That characterisation may or may not be accurate. The consumer concerns are real regardless.

The Octane Benefit Is Real — But Requires Hardware That Doesn't Exist Yet

The octane benefit of ethanol is sometimes invoked to challenge the mileage-loss narrative, and it deserves a fair hearing. Ethanol blends carry a blending RON of 120–135, substantially higher than base petrol. In an engine specifically designed and calibrated to exploit this octane number through a higher compression ratio, the efficiency gain can meaningfully offset the calorific penalty and even produce net efficiency improvements on a distance-per-litre basis. This is correct — for engines built after April 2025 to E20-tuned specifications. It is not correct for the several hundred million vehicles currently on Indian roads that were built before that date. The octane benefit requires dedicated hardware and calibration. It is not a feature that exists in the legacy fleet by default, and it cannot be retrofitted economically.

What Higher Blends Actually Require

Material compatibility compounds the picture at higher blends in ways that make E20 a genuine inflection point rather than just another step on a smooth ramp. Ethanol is hygroscopic — it absorbs water, promoting phase separation in fuel systems — and degrades the elastomers, rubber seals, and certain plastics used in pre-2023 fuel systems, while corroding aluminium components not designed for sustained ethanol exposure. These risks are manageable at E10 and broadly manageable at E20 with the correct material specifications. Moving from E20 to E85 is a fundamentally different engineering exercise. Dedicated flex-fuel engines require separate fuel maps, different injectors, reinforced fuel lines and seals, a cold-start fuel management strategy — ethanol's higher heat of vaporisation means substantially harder cold starts at high-blend ratios, a problem Brazil solved first with a secondary gasoline tank and later with Bosch's Flex Start fuel-heating technology — and substantively different ECU calibration across the entire operating map. These are new vehicles, not updated ones. The consumer absorbs the cost of that newness at 18–40% GST.

Exhibit 4 — fuel property comparison across E0, E10, E20, E85, and E100, covering

volumetric energy density, RON, stoichiometric air-fuel ratio, heat of vaporisation, and oxygen content — would frame why each step up the blend ladder is qualitatively more demanding than the last, and why the jump from E20 to E85 is not a continuation of the E5-to-E20 journey.

5. The World Has Tried This — And Stopped

The most important evidence in this note does not come from India. It comes from everywhere else that has tried to do what India's ethanol ambition now envisages. The record is consistent, it spans multiple continents and political systems, and its lesson is the same in every case: the blend wall is real, and only one country has broken through it.

The United States: Stuck at E10 Despite Every Structural Advantage

No country has invested more political and financial capital in ethanol than the United States, and no country has demonstrated more conclusively that high-blend ethanol — absent extraordinary structural conditions — simply does not achieve commercial scale. The US produces more ethanol than any nation on earth, running more than 15 billion gallons of corn-starch ethanol annually through a Renewable Fuel Standard framework sustained since 2005. The result: E10 is the de facto national standard, covering more than 95% of petrol sold. E15 is approved for vehicles manufactured after 2001 but has achieved only marginal market penetration. E85 is available at roughly 4,300 of approximately 150,000 petrol stations, serving around 21 million flex-fuel vehicles — but at sales volumes that are a rounding error of the total fuel market. The American Fuel and Petrochemical Manufacturers estimated that compelling meaningful E15 and E85 uptake through the RFS framework would cost approximately \$770 per incremental ethanol gallon.

The US proves something important. Even when ethanol is cheap, even when it is mandated, and even when the domestic political economy strongly favours it — the blend wall holds at approximately E10.

China and Thailand: Two Ways to Fail

China is instructive for India because the failure mechanism is structurally analogous. China announced a nationwide E10 mandate in 2017, targeting national implementation by 2020. By 2019-20, it had quietly begun unwinding the ambition — blend rates never exceeded approximately 4%, with later reports suggesting a softening to E5 as the effective target. The stated reason was food security: the government concluded that the feedstock requirements of a national ethanol mandate were in direct tension with its grain-reserve policy. India's feedstock pivot toward maize and rice creates exactly the same tension.

Thailand's case is the most recent and perhaps the most directly relevant, because Thailand

was the one country in the region that had genuinely constructed an E85 market. Through aggressive State Oil Fund subsidies, Thailand built out gasohol at E10, E20, and E85, achieving a blended rate of approximately 13.8% in 2018. Then the arithmetic of the subsidy caught up with the ambition. PTT and OR — Thailand's dominant downstream fuel companies — discontinued E85 sales from February 1, 2026. The full phase-out of E85 and gasohol 91/95 is scheduled for 2027, with consolidation on E20 and a diversion of surplus ethanol capacity — running at approximately 50% above road demand — toward Sustainable Aviation Fuel, where a 1% SAF mandate from January 2026 is rising to 8% by 2036. Thailand did not retreat from high-blend ethanol because it lacked conviction. It retreated because the subsidy cost was structurally unsustainable under the State Oil Fund Act.

Europe: The Electrification Bet

Europe has never seriously contemplated the high-blend path, and deserves brief mention primarily for what its policy architecture reveals about where institutional consensus has landed. RED III — the Renewable Energy Directive's third iteration — targets a 29% renewable transport share by 2030 but caps first-generation food-crop biofuels at 7%, frozen at 2020 consumption levels, and places a 5.5% sub-target on advanced fuels and renewable fuels of non-biological origin. The explicit intent is to crowd in electrification and green hydrogen, and to prevent crop-based biofuels from expanding at the expense of either food systems or more sophisticated decarbonisation pathways. E20 is only now being normalized in European standard-setting bodies, in 2026 — a decade after India set the same target.

Brazil: The Exception That Proves the Rule

Then there is Brazil — the only country that has genuinely broken through the E20 ceiling and sustained high blends over decades. Brazil moved to E30 from August 1, 2025, under CNPE Resolution 9/2025 and the Fuel of the Future Law 14.993/2024, which permits blend levels up to E35 on proof of technical viability. Approximately 90% of Brazil's light vehicle fleet — roughly 31.8 million cars and 7 million motorcycles — runs on flex-fuel technology capable of any blend from E0 to E100. Hydrous E100 is available at essentially every petrol station and prices in real time against gasoline, with consumers choosing on a roughly 70% price-parity rule: ethanol becomes economical when priced below 70% of gasoline per kilometre. In 2020, hydrous ethanol was priced at approximately 39% below gasoline. Sugarcane yields approximately 7,000 litres of ethanol per hectare, with an energy balance of 8–10 times, and bagasse cogeneration makes mills largely energy self-sufficient. The country set aside roughly 8 million hectares for sugarcane — approximately 1% of its total land area — without any meaningful food-security tension, on abundant, well-watered land.

Exhibit 5 — a comparison mapping ethanol blend levels achieved against key structural parameters (arable land availability, water stress, crop energy balance, flex-fleet penetration, consumer fuel choice) across Brazil, the US, EU, China, Thailand, and India —

would make the argument visually unmistakable. Brazil is a structural outlier in every dimension. India sits closer to China and Thailand than it does to São Paulo.

The lesson of the global experience is consistent. Advanced markets do not go to high blends because the blend wall is real: without dedicated flex-fuel vehicles, high-blend ethanol requires an infrastructure overhaul the economics do not support; without consumer fuel choice, the energy-density penalty is invisible to the market but borne by the consumer; without yield and water endowments comparable to Brazil, feedstock economics deteriorate rapidly; and everywhere except Brazil, electrification has emerged as the deeper, more structurally durable decarbonisation bet. The world did not fail to reach high blends through lack of trying. It stopped trying because the evidence accumulated.

6. Flex-Fuel: Why the Policy Vision Will Not Become the Commercial Reality

The policy narrative around flex-fuel in India has accelerated sharply in 2026. Maruti Suzuki launched the Wagon R FFV — E20 to E100 capable, homologated for E85 — on June 4, 2026. Hero MotoCorp launched flex-fuel variants of the Splendor+ and HF Deluxe on June 3, 2026; TVS and Bajaj have flex-fuel two-wheelers already in market. Petroleum Minister Hardeep Singh Puri, addressing the Maruti launch, cited a target of 500 E85 retail stations by December 2026 and 5,000 by end-2027, and claimed a three-year payback on the flex-fuel vehicle premium at E85 pricing. The government's communication is unambiguous: flex-fuel is the bridge from E20 to the high-blend future.

We believe the commercial reality will disappoint the policy ambition — and for reasons that are structural rather than temporary.

The GST Architecture Is the Most Honest Policy Signal Available

The GST treatment of vehicles is the clearest expression of the actual policy hierarchy. Flex-fuel vehicles, alongside conventional ICE and strong hybrids, attract GST at 18–40% depending on segment and engine size. Battery electric vehicles attract 5%. This differential was not set by accident. It reflects a revealed policy preference for electrification as the long-run transport decarbonisation pathway, with ethanol-ICE and flex-fuel positioned as transitional bridges at best. The Indian automobile GST framework does not send a price signal that says “buy flex-fuel.” It sends a price signal that says “buy EVs if you can, accept ICE in the meantime.” Until this changes — and there is no firm indication that it will, despite persistent industry lobbying from ISMA and SIAM — the economics of the flex-fuel premium are working against the adoption they are supposed to enable.

The Infrastructure Gap Is More Fundamental Than It Appears

The E85 retail infrastructure problem is more fundamental still. Flex-fuel vehicles are commercially meaningful only if the consumer has access to the fuel that makes them worthwhile — E85, at a material price discount to E20, at enough stations that range anxiety does not dominate the purchase decision. As of mid-2026, E85 retail availability is essentially nascent outside a handful of pilot installations. The government's target of 500 stations by December 2026 is a starting point, not a network. Compare this to Brazil's flex-fuel transition: E100 was available at essentially every station when FFV sales began in earnest in 2003 — because it had been sold for two decades before the FFV arrived. India does not have that foundation, and the chicken-and-egg problem between vehicle adoption and fuel retail cannot be resolved by government targets alone.

Even the economics of the vehicle premium require scrutiny. Minister Puri's claimed three-year payback on the flex-fuel vehicle cost at E85 pricing works only under a specific and optimistic set of assumptions: that E85 is priced at a substantial and sustained discount to E20 per kilometre, that the consumer drives enough to realise the running-cost saving within a reasonable period, and that the vehicle is purchased in a city with material E85 availability. If E85 is priced at parity with E20 per litre — which, given the administered pricing system and the energy-density differential, would actually represent a per-kilometre penalty for the consumer — the payback case collapses. Maruti itself has stated that meaningful flex-fuel volume is five to ten years away. That is not the language of a company expecting a near-term commercial transition. It is the language of a company complying with a policy requirement while managing expectations responsibly.

Two-Wheelers Are a Partial Exception — But Face Their Own Competitive Threat

The two-wheeler flex-fuel market deserves its own framing and is a genuine, if partial, exception to our cautious view. India's two-wheeler segment accounts for roughly 61% of all petrol consumption — 61.4% on the PPAC/Nielsen study — making it the single most important battleground for the ethanol programme. Flex-fuel motorcycles are in market, the incremental cost is lower than in passenger vehicles, and the rural and semi-urban use case — where range patterns suit higher-blend spot availability — is more favourable than the urban passenger vehicle market. Hero, TVS, Bajaj, and Suzuki all have products in place.

But the two-wheeler segment also faces the sharpest EV competition. Battery technology and charging logistics for two-wheelers are simpler than for passenger cars. Total cost of ownership for EVs in the segment is already competitive in urban markets on a well-to-wheel basis, and government incentives have meaningfully accelerated urban adoption. The flex-fuel motorcycle opportunity is real; it is not certain to survive the two-wheeler EV transition. The question — for Hero, Bajaj, and TVS, whose downstream segment exposure is highest — is which transition happens faster. We think EVs win urban two-wheelers within five years; flex-fuel motorcycles have a better near-to-medium term case in rural markets, where charging infrastructure is thinner and fuel-station availability at the local pump

remains more reliable than home or public charging.

7. The Alternatives Are Not Waiting: Hybrids, EVs, and the Structural Limits of What Ethanol Can Actually Address

The case for higher-blend ethanol rests implicitly on a proposition: that in the absence of a compelling alternative, ethanol is the best available decarbonisation lever for India's road transport sector. That proposition deserves testing against what the alternatives actually offer.

Strong Hybrids: The Most Underrated Vehicle in India's Transition

Strong hybrids are the most immediate competitive alternative in the passenger vehicle segment, and they receive less analytical attention than they deserve. A strong HEV — not a mild hybrid, not a 48-volt belt-starter-generator — delivers real-world fuel efficiency improvements of 25–40% over a comparable ICE engine in Indian driving conditions: start-stop traffic, moderate highway speeds, the operating environment that describes most Indian passenger vehicle use. No range anxiety. No charging infrastructure dependency. No incremental consumer behaviour change required. The fuel it runs on is E20 petrol — the infrastructure is already built.

The value proposition of the strong hybrid is, in effect, a 25–40% reduction in petrol consumption per kilometre, implemented immediately, at scale, without waiting for a new fuel distribution network or a new consumer behaviour around refuelling. That is a more powerful near-term decarbonisation intervention than incrementally higher ethanol blends in a petrol pool that represents only 40 MMT of a 239 MMT petroleum consumption base.

The obstacle to strong hybrid scaling is GST, not technology. Strong hybrids currently sit at 28% plus cess — the same bracket as large diesel SUVs — against 5% for EVs and 18–28% for mass-market ICE. This positions strong hybrids as a premium-segment product when their efficiency benefit is most valuable in the mass market. Toyota and Maruti are the primary current beneficiaries; rationalisation of the GST to say, 12–15%, would be a more powerful automotive-sector decarbonisation lever than any achievable incremental ethanol blend progression in the near term.

Battery EVs: Winning in Segments, Not All at Once

Battery electric vehicles present a segmented picture, and it is worth resisting both the extreme bull and extreme bear interpretations. In two-wheelers and three-wheelers, the EV case is already strong on total cost of ownership in urban markets — daily range requirements are modest, home or workplace charging eliminates range anxiety for most

use patterns, energy costs per kilometre are compelling, and the FAME programme plus state subsidies have compressed the price premium substantially. ICCT's 2021 lifecycle analysis found Indian BEV lifetime emissions approximately 19–34% lower than comparable gasoline cars on the current coal-heavy grid, with the advantage growing as the grid greens. India's installed solar and wind capacity has crossed 200 GW and is expanding rapidly.

In passenger vehicles, the EV case is trajectory-dependent rather than immediate. Range anxiety, charging infrastructure density outside major cities, and residual value uncertainty remain real constraints. But the trajectory is one-directional: each year, the infrastructure improves, prices fall, and the range extends. The question for passenger vehicles is not whether EVs win; it is when. We believe that horizon is post-2028 for meaningful mass-market penetration, with premium and urban segments moving earlier.

The Lifecycle Carbon Complication

There is a lifecycle carbon dimension to this comparison that ethanol advocates rarely confront directly, and it has become more uncomfortable as the feedstock mix has shifted. Sugarcane-based ethanol does deliver genuine lifecycle CO₂ reductions — the US EPA's 2010 RFS2 analysis found Brazilian sugarcane ethanol reduces GHG emissions by approximately 61% versus gasoline on a 30-year ILUC-adjusted basis, and India's own sugarcane ethanol likely achieves reductions in a similar range. But India's ethanol is no longer primarily sugarcane-based. Maize ethanol carries approximately double the lifecycle GHG intensity of sugarcane ethanol; rice-based ethanol is worse still. As the feedstock mix has shifted toward grain, the programme's carbon credentials have shifted with it — and the headline claim of 909 lakh MT of CO₂ avoided, cited by Minister Puri in June 2026, is a blended figure whose composition, on current feedstock trends, is less impressive than the number suggests.

The Diesel Boundary: Ethanol's Structural Ceiling

There is one dimension of this story that is consistently underweighted in the programme's advocacy and deserves to be stated plainly. Ethanol is a petrol substitute. It has no equivalent chemistry or logistics pathway into the diesel fuel system. Diesel accounts for approximately 91 MMT of India's petroleum consumption — against petrol's 40 MMT and a total petroleum base of 239 MMT. Even total success in the petrol pool — every litre of petrol replaced by ethanol, E100 fleet, Brazil-equivalent endowments and economics — would leave the diesel problem entirely untouched.

Exhibit 6 — a single bar chart showing petrol (40 MMT), diesel (91 MMT), and total petroleum consumption (239 MMT), with a shaded region showing ethanol's maximum theoretical addressable market — would make this structural ceiling immediate and visual. Ethanol's ambition is bounded by the petrol pool. The headline crude import dependence number — 88.2%, roughly \$137 billion gross in FY2024-25 — will not materially change

regardless of what ethanol achieves in the petrol segment, because the diesel dependency runs in parallel and is, for now, beyond ethanol's reach.

8. What This Means for Autos: The Winners Are Already Clear

The automotive sector's relationship with India's ethanol programme is frequently described in terms of transition risk — the risk that an accelerating ethanol mandate requires costly vehicle modifications, disrupts existing model architectures, or disadvantages OEMs that have bet on EVs over ICE. We think that framing misses the point. The E20 transition is already complete. OEMs have made the necessary engineering adjustments; the compliance timeline has been met; material and calibration upgrades are embedded in new vehicles from April 2025 onward. The question is not whether the auto sector can manage the E20 transition — it has managed it. The question is whether there is a further transition coming that requires a different strategic response, and our answer is: not at commercial scale, not within the investment horizon that matters.

E20-Calibrated ICE and Strong Hybrids: The Structural Winners

The structural winner in the near-to-medium term is the combination of E20-calibrated ICE and the strong hybrid. Both run on the existing fuel infrastructure. Neither requires a consumer to make a bet on a fuel that may or may not be available at their nearest pump. Both benefit from the same engine engineering that OEMs have already invested in calibrating for E20 performance — the octane benefit of higher-blend petrol is now captured in dedicated hardware for new vehicles, delivering efficiency improvements that the legacy fleet could not access. OEMs with strong positions in both ICE and hybrid — Toyota, Maruti, Hyundai — are well-positioned regardless of how the higher-blend story develops, and their competitive position actually improves if flex-fuel fails to scale, because the alternative market share goes to hybrids and E20-ICE, not to the margins.

Flex-Fuel: A Regulatory Option, Not a Commercial Transition

The flex-fuel vehicle opportunity is real but bounded. Exhibit 7 — a total cost of ownership comparison across E20-ICE, flex-fuel FFV (at varying E85 price assumptions), strong HEV, and BEV for a mass-market passenger vehicle — would illustrate why the FFV's TCO advantage over E20-ICE is highly sensitive to E85 pricing and availability, and requires sustained E85 discounts of more than 25–30% per kilometre to justify the vehicle premium at current GST rates.

Flex-fuel will scale in two-wheelers before passenger vehicles — the cost differential is lower, rural range requirements suit the available infrastructure, and the competitive threat from EVs is slower outside urban markets. In passenger vehicles, flex-fuel is best

understood as a regulatory compliance vehicle: present in OEM portfolios because the government has asked for it, scaling slowly because the economics of E85 are not yet in place, and ultimately subordinate to the EV trajectory in the medium term. OEMs that have invested in flex-fuel capability are not wasting the investment — they are buying a regulatory option that protects market position if the government accelerates the mandate. But modelling it as a commercial transition driver within five years would be a mistake.

The Signals That Would Change Our View

The signal to watch — and the one that would cause us to upgrade our view on higher blends — is not a ministerial speech or a BIS notification. Notifications are policy signalling; they do not move fuel. Three material conditions would genuinely change the commercial outlook. A firm, time-bound government notification mandating E25–E27 retail delivery after October 2026 — not just regulatory specification, but a rollout timeline with OMC accountability — would be the first. GST rationalisation for flex-fuel vehicles to something approaching EV parity would be the second; this is the most powerful lever available and the most politically difficult. And E85 priced consistently at a discount sufficient to make the vehicle-level payback credible at 500 or more stations outside metro areas would be the third. None of those three conditions is confirmed as of mid-2026. Until they are, the flex-fuel narrative remains an option being positioned, not a transition underway.

Conclusion: The Blend Wall Is Real, E20 Is the Destination, and the Auto Industry Should Position Accordingly

India's ethanol programme is a genuine policy success story, and it deserves to be recognised as one. Reaching E20 — six years ahead of schedule, at scale, across a country of 1.4 billion people and several hundred million vehicles — is a logistics and policy achievement that most countries never approach. The energy-security benefit, the farm-income transfer, and the rough infrastructure proof-of-concept are real, even if the precise benefit figures from government are not independently audited and should be read with appropriate caution.

But the programme has peaked. The economics have inverted. The feedstock has migrated from by-product to food crop. The legacy fleet has hit its compatibility ceiling. The global precedent is consistent and clear. And the conditions for a flex-fuel market — GST parity, E85 infrastructure, a consumer price incentive — are not in place. A slow administrative progression toward E25–E27 over the rest of this decade is the most likely path; E85 and E100 at mass adoption within ten years is not.

The auto industry analogy that fits India's situation best is not Brazil in 2003, when the first FFV launched into a market that already had hydrous E100 at essentially every station. It is

the United States in 2007, when E10 became universal and a decade-long debate about E15 and E85 began. They have not followed, despite the most aggressive biofuel policy framework in the world outside Brazil. India's version of that debate has only just begun — and the answer, we believe, will rhyme with America's more than it will with Curitiba's.

For investors, the conclusion is direct. E20-calibrated ICE and strong hybrids are the structural winners through the end of the decade. Flex-fuel is a watched option, not a confirmed transition. EVs are winning two-wheelers and will win passenger vehicles with time. And ethanol — even at its most ambitious — will not touch the diesel pool. India's crude import dependence is not an ethanol problem. It is a much bigger problem than that, and the solutions to the bigger problem are being built somewhere else entirely.

Data notes: All PPAC petroleum consumption data is for FY2024-25 unless noted. Ethanol supply year (ESY) runs October–September. Government benefit claims for forex savings, CO2 avoided, and farmer income are MoPNG self-stated figures that have escalated across successive ministerial statements; these are not independently audited and an independent estimate of crude savings is materially lower. The \$137.2 billion gross crude import bill (PPAC) and ICRA's net oil import estimate of approximately \$101–104 billion are not directly comparable metrics — the former is gross crude-only at actuals; the latter nets out product exports and is based on assumed crude prices, not actuals. LocalCircles consumer survey data on mileage loss and vehicle wear is self-reported online-survey data, not instrumented testing; the figures are cited alongside official ARAI estimates for context, not as engineering evidence. Feedstock water-intensity figures source to NITI Aayog 2021 Roadmap and University of Twente data; CSTEP for irrigation demand projections. Flex-fuel vehicle launch details as reported by OEMs and PIB as of mid-2026; E85 retail availability remains nascent. All exhibit descriptions throughout this note specify the argumentative conclusion each exhibit should illustrate; the underlying datasets are tabulated separately.